Published on *Statistics* (https://onlinecourses.science.psu.edu/statprogram)

2.0 - Matrix Arithmetic

To take the **transpose** of a matrix, simply switch the rows and column of a matrix. The transpose of A can be denoted as A' or A^T. For example, for

$$A = \begin{pmatrix} 1 & -5 & 4 \\ 2 & 5 & 3 \end{pmatrix}$$

$$A' = A^T = \begin{pmatrix} 1 & 2 \\ -5 & 5 \\ 4 & 3 \end{pmatrix}$$

If a matrix is its own transpose, then that matrix is said to be **symmetric.** Symmetric matrices must be square matrices, with the same number of rows and columns. One example of a symmetric matrix is shown below:

$$A = \begin{pmatrix} 1 & -5 & 4 \\ -5 & 7 & 3 \\ 4 & 3 & 3 \end{pmatrix} = A' = A^T$$

To perform matrix **addition**, two matrices must have the same dimensions. This means they must have the same number of rows and columns. In that case simply add each individual components, like below.

$$A + B = \begin{pmatrix} 1 & -5 & 4 \\ 2 & 5 & 3 \end{pmatrix} + \begin{pmatrix} 8 & -3 & -4 \\ 4 & -2 & 9 \end{pmatrix} = \begin{pmatrix} 1+8 & -5-3 & 4-4 \\ 2+4 & 5-2 & 3+9 \end{pmatrix} = \begin{pmatrix} 9 & -8 & 0 \\ 6 & 3 & 12 \end{pmatrix}$$

Matrix addition does have many of the same properties as "normal" addition, such as

$$A + B = B + A$$

$$A + (B + C) = (A + B) + C$$

In addition, if one wishes to take the transpose of the sum of two matrices, then

$$A^T + B^T = (A + B)^T$$

To multiply a matrix by a scalar, also known as **scalar multiplication**, multiply every element in the matrix by the scalar. For example:

$$6 * A = 6 * \begin{pmatrix} 1 & -5 & 4 \\ 2 & 5 & 3 \end{pmatrix} = \begin{pmatrix} 6*1 & 6*-5 & 6*4 \\ 6*2 & 6*5 & 6*3 \end{pmatrix} = \begin{pmatrix} 6 & -30 & 24 \\ 12 & 30 & 18 \end{pmatrix}$$

To multiply two vectors with the same length together is to take the **dot product**, also called **inner product**. This is done by multiplying every entry in the two vectors together and then adding all the products up. For example, for vectors *x* and *y*, the dot product is calculated below

$$x \cdot y = \begin{pmatrix} 1 & -5 & 4 \end{pmatrix} * \begin{pmatrix} 4 & -2 & 5 \end{pmatrix} = 1 * 4 + (-5) * (-2) + 4 * 5 = 4 + 10 + 20 = 34$$

To perform **matrix multiplication**, the first matrix must have the same number of columns as the second matrix has rows. The number of rows of the resulting matrix equals the number of rows of the first matrix, and the number of columns of the resulting matrix equals the number of columns of the second matrix. So a 3 × 5 matrix could be multiplied by a 5 × 7 matrix, forming a 3 × 7 matrix, but one cannot multiply a 2 × 8 matrix with a 4 × 2 matrix. To find the entries in the resulting matrix, simply take the dot product of the corresponding row of the first matrix and the corresponding column of the second matrix. For example:

$$C * D = \begin{pmatrix} 3 & -9 & -8 \\ 2 & 4 & 3 \end{pmatrix} * \begin{pmatrix} 7 & -3 \\ -2 & 3 \\ 6 & 2 \end{pmatrix}$$

$$C * D = \begin{pmatrix} 3 * 7 + (-9) * (-2) + (-8) * 6 & 3 * (-3) + (-9) * 3 + (-8) * 2 \\ 2 * 7 + 4 * (-2) + 3 * 6 & 2 * (-3) + 4 * 3 + 3 * 2 \end{pmatrix}$$

$$C * D = \begin{pmatrix} 21 + 18 - 48 & -9 - 27 - 16 \\ 14 - 8 + 18 & -6 + 12 + 6 \end{pmatrix} = \begin{pmatrix} -9 & -52 \\ 24 & 12 \end{pmatrix}$$

Matrix multiplication has some of the same properties as "normal" multiplication , such as

$$A(BC) = (AB)C$$

$$A(B + C) = AB + AC$$

$$(A + B)C = AC + BC$$

However matrix multiplication is not communicative. That is to say *A*B* does not necessarily equal *B*A*. In fact, *B*A* often has no meaning since the dimensions rarely match up. However, you can take the transpose of matrix multiplication. In that case $(AB)^T = B^T A^T$.

Source URL: https://onlinecourses.science.psu.edu/statprogram/node/149